PRABHAT SINGH

Email | Phone | LinkedIn | GitHub | X

SKILLS

- **Languages & Backend:** Python, asyncio, FastAPI, Websockets, PostgreSQL, SQLAlchemy, Pydantic
- **Agentic AI & Memory:** Agent loop design,tool/function calling, agent memory (short-term and long-term), context management, RAG, prompt engineering, Mem0, MCP, self-improvement
- **Voice AI :** STT/LLM/TTS pipelines, real-time streaming, LiveKit (WebRTC), Deepgram Flux and Aura-2, Cerebras and Groq inference
- **LLMs & Vector Databases:** Gemini, Qwen, GPT-OSS, ChromaDB
- **Security & Systems :** AES-256 encryption, FastAPI , RBAC, Active Directory/LDAP, Nextcloud API, file system monitoring
- **Tools & Platforms :** Git, GitHub, Docker, Linux, Claude Code, Postman, pytest

PROJECTS

Voice Scheduling Agent | Python, asyncio, LiveKit, Deepgram, Cerebras, Google Calendar Ava-github

- Built Ava, a real-time voice scheduling agent from scratch, without an agent framework. It runs a streaming STT → LLM (tool calling) → TTS pipeline over WebRTC and books, reschedules and cancels Google Calendar appointments, achieving approximately 0.6–0.9 seconds of end-to-end latency.
- Prevented hallucinated bookings using a code-level harness: dynamic date-aware system prompt, directive tool schemas, per-turn verification of successful tool calls, and graceful tool-error handling. Integrated Mem0 to store long-term user facts across sessions and load them at session start .
- Implemented local CPU-based Silero VAD with 32 ms windowing, exponential smoothing, hysteresis thresholds and a start/end state machine, achieving about 0.5 ms of inference latency per frame. Validated the implementation using deterministic state-machine tests and live-mic tests.

Coding Agent Harness | Python, OpenRouter, Mem0, MCP Engram-github

- Built Engram, a lightweight terminal-based coding agent that streams LLM responses and executes tools for reading, writing, editing, and running shell commands through a custom agent loop, without external agent-framework dependencies.
- Used OpenRouter as a provider-agnostic model layer, enabling model switching without code changes.
- Integrated GitHub through MCP, allowing the agent to work with existing codebases: browsing repositories, reading and editing files, and managing issues and pull requests.
- Integrated Mem0 for persistent memory, enabling the agent to retain project context and user preferences across sessions.

EXPERIENCE

Software Developer

WESEE (Research and development lab) | May 2024 - August 2025

- Built CloudGuard, Fastapi based web application integrated with Nextcloud APIs for secure file uploads, downloads, synchronization and sharing across local and cloud storage.
- Implemented AES-256 file encryption with custom key management and integrated Active Directory/LDAP authentication with role based access control(RBAC), ensuring that only encrypted files were stored in Nextcloud and decryption was restricted to authorized users.
- Developed a watchdog-based folder monitor that automatically encrypts newly added files, along with double-click-triggered decryption and timed deletion of decrypted files.

EDUCATION

- **Super30 Program | 100xSchool | May 2026 – November 2026**
  Selective, cohort-based program focused on building AI/Agentic systems
- **M.Sc in Computer Science (AI and Machine Learning)**
  Scaler Neovarsity | Expected 2026
- **M.Tech in Aerospace (Flight Mechanics and Control)**
  IIT Kanpur | 2022
- **B.Tech in Electrical and Electronics Engineering**
  Galgotias College of Engineering and Technology | 2018